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Remote Locations Enhanced Safety Using Drone Technology

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Abstract

Drones, also referred to as unmanned aerial vehicles, or UAVs, have shown a growing potential for enhancing operational safety in high-risk industries. A low-cost, embedded UAV platform with environmental sensors and thermal imaging is presented in this paper for proactive safety monitoring at distant oil and gas locations. The system conducts autonomous inspections to identify early-stage anomalies prior to equipment start-up or human entry by combining sensors like flame, gas, smoke, humidity, temperature, and distance modules with an Arduino microcontroller. The platform's sensitivity to environmental risks, such as gas leaks and overheating, was confirmed by controlled field tests. The suggested system minimizes manual intervention and allows early risk detection because it runs continuously, unlike traditional survey drones. According to the findings, operational safety is greatly improved, downtime is decreased, and a foundation is provided by real-time drone-based monitoring for future AI-enabled inspection systems.

Introduction

Remote and offshore oil and gas sites pose significant challenges for routine inspection due to limited accessibility, harsh environments, and elevated safety risks. Traditional manual inspections often result in delayed fault detection and increased personnel exposure, potentially leading to undetected leaks or equipment failures.

This work presents a novel implementation of UAVs as embedded, autonomous agents in daily safety operations. Each UAV is permanently stationed at a designated remote or offshore site and programmed with inspection routines tailored to the local layout, operational priorities, and risk factors. Triggered by events such as equipment startup, alarm conditions, or adverse weather, the UAV performs autonomous flights to monitor for hazards using onboard environmental sensors.

The system enables early detection of anomalies—such as heat spikes, gas presence, or fluid leaks—supporting timely interventions and reducing operational risks. This paper outlines the system architecture, sensor integration, and experimental validation, highlighting its potential for real-time safety monitoring and cost-effective deployment in offshore environments.

Literature Review

Despite rapid technological progress and a growing number of successful UAV deployments in the oil and gas sector, most applications remain fragmented, task-specific, and lack full autonomy. Current UAV inspection systems often rely on manual piloting and post-flight data interpretation, requiring skilled operators to navigate and analyze sensor outputs. This approach is time-consuming and introduces potential for human error, as noted by [Aromoye, Olowu, and Rezaei \(2025\)](#).

Few existing UAV platforms are designed to autonomously perform repeated, multi-modal safety inspections. Most systems lack onboard processing capabilities needed for real-time anomaly detection and adaptive response, limiting their effectiveness for proactive safety management in dynamic offshore environments ([Aromoye et al., 2025](#)). This gap is particularly critical in applications requiring early detection of hazards such as gas leaks, overheating components, or structural degradation.

In response, this work proposes a multi-sensor UAV platform capable of executing routine inspection missions independently. The system integrates onboard environmental sensing, local processing, and pre-programmed routines tailored to offshore layouts, enabling real-time hazard recognition with reduced reliance on ground-based control.

Objectives/Scope

This work aims to develop and evaluate a low-cost, multi-sensor safety monitoring system for autonomous UAVs or fixed stations operating in offshore oil and gas environments. The system integrates environmental sensors into a unified embedded platform to detect hazardous events such as fire, gas leaks, or abnormal environmental conditions. The objective is to assess the feasibility of real-time sensing and alert generation using compact, Arduino-compatible hardware. Validation is conducted through laboratory experiments simulating field-relevant hazards, while synthetic data is used to replicate UAV-based monitoring scenarios. This platform is intended to support integration with SCADA systems and AI models for autonomous offshore safety decision-making.

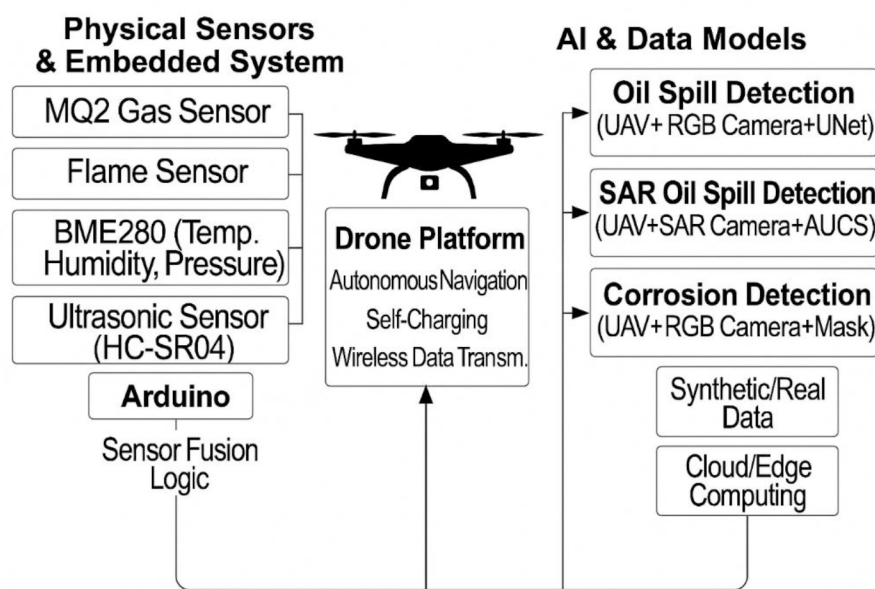


Figure 1—SYSTEM DIAGRAM OF AN AUTONOMOUS DRONE PLATFORM FOR ROUTINE INSPECTION INTEGRATING PHYSICAL SENSORS, SENSOR FUSION, AND AI MODELS FOR OIL SPILL AND CORROSION DETECTION, WITH CLOUD/EDGE COMPUTING FOR REAL-TIME DATA PROCESSING AND FEEDBACK.

Theoretical Background

The proposed safety monitoring system is grounded in a dual development strategy that combines experimental prototyping with synthetic modeling. Experimental validation focuses on the real-time behavior of the embedded system, evaluating sensor response accuracy, event detection logic, and alert triggering under controlled conditions. In parallel, synthetic simulations are used to abstract inspection workflows and assess system performance in offshore-like scenarios without the constraints of field deployment. This complementary approach enables both technical validation of the sensing logic and early-stage system refinement, contributing to a robust foundation for future autonomous UAV operations in offshore environments.

Experimental Sensor-Based Components

This section describes the physical sensors integrated into an Arduino-based prototype and validated through controlled laboratory trials simulating offshore safety scenarios such as gas leaks, flame presence, proximity alerts, and thermal anomalies. Each sensor is selected based on its operating principle and relevance to real-time hazard detection in UAV-based offshore monitoring.

Core Sensor Modules

The following subsections describe the major prototype components, including their theoretical operating principles and roles in the system.

Ultrasonic Sensor (HC-SR04) - Proximity Detection. The HC-SR04 ultrasonic sensor was integrated into the UAV system to enable low-altitude proximity sensing and obstacle detection around offshore infrastructure. Its ability to measure short-range distances enhanced collision avoidance during close inspections and supported safe autonomous landings on constrained surfaces.

The sensor uses a time-of-flight method to calculate distance by emitting an ultrasonic pulse and measuring the time it takes for the echo to return. The distance to an object is given by:

$$Distance(cm) = \frac{v \cdot t}{2}$$

Where:

v is the speed of sound in air (approximately 343 m/s at 20 °C)

t is the total time taken for the pulse to travel to the object and back.

This real-time distance measurement capability also allows for monitoring personnel proximity or detecting obstructions near hazardous zones during UAV-assisted inspections.

BME280 Sensor - Temperature, Humidity, and Pressure. The BME280 sensor was integrated into the UAV system to provide real-time environmental monitoring during remote site inspections. It combines three sensing elements in a compact digital package:

- Thermistor-based temperature sensing
- Capacitive humidity sensing
- Piezoresistive barometric pressure sensing

These environmental parameters are critical for identifying early-stage hazards such as electrical overheating, moisture ingress in transformers or enclosures, and atmospheric pressure shifts associated with gas release events or thermal anomalies.

The barometric pressure readings also support altitude estimation for low-level UAV operations using the hypsometric equation:

The BME280 is a digital sensor integrating:

- Thermistor-based temperature sensing.
- Capacitive humidity sensing.
- Piezoresistive barometric pressure measurement.

These variables are affected during fires, gas releases, or heat leaks. The pressure data can be converted into altitude using the hypsometric equation:

$$h = 44330 \cdot \left(1 - \left(\frac{P}{P_0}\right)^{1/5.255}\right)$$

Where:

- h = estimated altitude (m)
- P = measured pressure (hPa),.
- P_0 = standard sea-level pressure (1013.25 hPa).
- This capability enables the UAV to maintain accurate positioning near elevated equipment and improves stability during close-range inspections around flare stacks, transformers, and flowlines.

By detecting, the BME280 enhances the system's ability to identify developing faults, such as heat buildup, internal condensation, or environmental trends linked to equipment degradation. It serves as a complementary sensor to thermal and gas detectors, contributing to multi-layered anomaly detection in hazardous offshore and remote environments.

MQ2 Sensor - Smoke and Gas Detection. The MQ2 gas sensor, a metal oxide semiconductor (MOS)-based device, was used to detect the presence of volatile gases and smoke, including hydrogen sulfide (H_2S). While dedicated H_2S sensors are typically installed at wellsite, they may occasionally generate false alarms. Therefore, the MQ2 sensor was deployed during UAV- based inspections to routinely monitor for potential gas leaks, verify H_2S presence, and help localize emission sources for confirmation.

The MQ2 gas sensor is based on a metal oxide semiconductor (MOS). When gases like methane, butane, LPG, and smoke are present, the sensing layer (SnO_2) alters its resistance:

$$R_s = \frac{V_{RL} \cdot R_L}{V_c - V_{RL}}$$

Where:

- R_s = sensor resistance in target gas
- V_{RL} = voltage across load resistor
- R_L = load resistance.
- V_c = circuit voltage.

By comparing $\frac{R_s}{R_0}$ (baseline in clean air) with known sensitivity curves, qualitative gas detection is achieved.

Flame Sensor - Infrared Flame Detection. An infrared flame sensor was integrated into the UAV payload to detect flame emissions in the 760–1100 nm range. This facilitated remote inspection of offshore flare stacks by verifying flame presence and stability, as well as identifying potential ignition hazards during gas leak scenarios. Its high sensitivity to hydrocarbon combustion signatures made it particularly valuable for safety monitoring in offshore environments.

The sensor operates using an infrared photodiode or phototransistor tuned to the same wavelength range (760–1100 nm), where open flames emit peak infrared radiation. When the detected IR intensity exceeds a predefined threshold, the sensor outputs a digital signal:

$$\begin{aligned} \text{Flame Detected} &= \{1, \text{ if } IR \text{ intensity} > \text{threshold}\} \\ &= \{0, \text{ otherwise}\} \end{aligned}$$

This rapid detection enables fast preemptive shutdowns or alerts in the presence of uncontrolled flames.

Sensor Fusion and Alert Logic. To support autonomous safety monitoring in remote and offshore locations, a simple yet effective sensor fusion algorithm was implemented. The logic is based on a compound Boolean expression that triggers an alert when environmental or hazard indicators deviate from safe thresholds:

$$\text{Alert} = ((T - T_0 > 10) \vee (H - H_0 > 10) \vee (D < 10) \vee (G = 0) \vee (F = 1))$$

Where:

- T, H = current temperature and humidity
- T_0, H_0 = baseline temperature and humidity
- D = ultrasonic distance (cm)
- G = gas digital output (0 = gas/smoke detected)
- F = flame detection flag (1 = flame detected)

This logic enables the UAV to autonomously identify abnormal environmental conditions, gas leaks, close-proximity hazards, or flame activity, triggering onboard alerts (e.g., buzzer) or transmitting data for real-time operator response.

By integrating data from multiple low-cost sensors, this system replicates essential functions of more complex industrial-grade monitoring solutions—while remaining lightweight and cost-effective. This multi-sensor synergy enhances early detection of safety risks before visible cues appear on thermal imaging or before advanced AI-based diagnostics are engaged in future system upgrades.

Methods, Procedures, and Process

Experimental Setup and Procedures. An Arduino Uno R3 microcontroller was connected to each sensor, and Serial Monitor was used to log data. To replicate safety-relevant scenarios like flame proximity, smoke exposure, gas presence, temperature fluctuation, and object proximity, controlled experiments were conducted. These comprised:

- Flame Test: going up to the flame sensor with a lit match or lighter.
- Smoke Detection: Seeing analog/digital spikes after lighting and extinguishing a match.
- Gas Simulation: To mimic the detection of volatile organic compounds (VOCs), place alcohol pads close to the MQ2 sensor.
- Ultrasonic Distance: A hand moved close to the sensor, mimicking an operator or human approaching.
- Environmental Shifts: Using the BME280, hot air is blown to replicate a sudden rise in temperature and humidity.

For traceability and multi-sensor correlation, each reading was timestamped.

Experimental Setup and Procedures. For 10–20 seconds per experiment, sensor outputs were recorded every 2 seconds. Among the data fields were:

- Timestamp
- Distance (cm)
- Smoke Analog & Digital
- Flame Detection (0/1)
- Temperature (°C)
- Humidity (%)
- Pressure (hPa)

A snapshot of the surroundings during a simulated event is represented by each row in the log. Tasks involving time-series prediction and classification can benefit from this data.

Preprocessing Steps. To prepare the data for downstream analysis and machine learning model development, the following preprocessing pipeline was used:

- Handling Missing Values: Synthetic data was complete; in real deployment, missing values (e.g., from sensor dropouts) will be forward-filled or imputed using interpolation methods.
- Feature Scaling: Continuous variables such as temperature, humidity, pressure, and distance were normalized using Min-Max Scaling to ensure equal model weighting:

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

- One-Hot Encoding: Binary categorical features (e.g., Flame Detected: 0/1) were preserved in native form. In future iterations with multi-class outputs, one-hot encoding will be applied.

Model Inputs.

- X (Features):
 - Distance
 - Smoke Analog
 - Smoke Digital
 - Flame
 - Temp
 - Humidity
 - Pressure
- y (Target):
 - Alert (1 if any of the conditions met: flame, smoke, gas, temp/humidity spike, object proximity)

Machine Learning Models. For proof-of-concept, the following models were evaluated using the synthetic dataset:

- Logistic Regression

- Decision Tree Classifier
- Random Forest Classifier
- K-Nearest Neighbors (KNN)

These models were selected for their interpretability and suitability for embedded/low-compute environments.

Cross-Validation and Hyperparameter Tuning.

- Cross-Validation: 5-fold cross-validation was used to ensure robust performance evaluation.
- Hyperparameters: Tree depth, number of estimators, and distance metrics were tuned using grid search.

Evaluation Metrics. Model performance was evaluated using standard classification metrics:

- Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- Mean Absolute Error (MAE):

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

- R-Squared (R^2):

$$R^2 = \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

While R^2 is typically used in regression tasks, it provided an interpretive baseline when applying logistic regression models on alert probability scores

Results and Observations

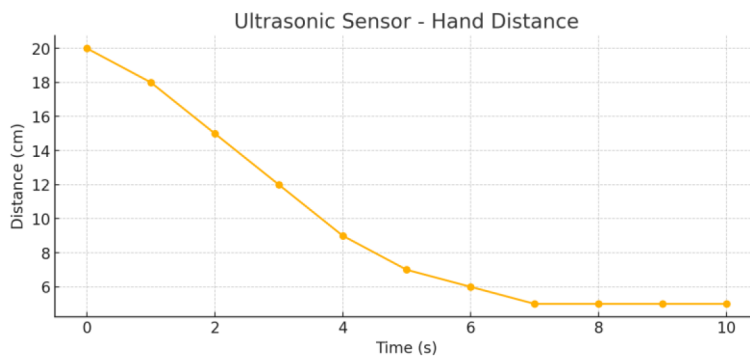
To evaluate the multi-sensor monitoring platform integrated with the Arduino Uno R3, four distinct experiments were performed. Each was designed to simulate realistic hazards or operational changes typically encountered at remote oil and gas sites. While real sensor readings were collected, synthetic data was used to simulate extended runs and support graphical analysis. The results validate that low-cost sensors can collaboratively detect environmental hazards when processed together.

Ultrasonic Proximity Detection. Test Setup: A human hand was gradually moved toward the ultrasonic sensor over a 10-second interval.

Observation:

The distance value dropped from ~100 cm to <10 cm.

Trigger conditions activated when the object was closer than 10 cm.



Result:

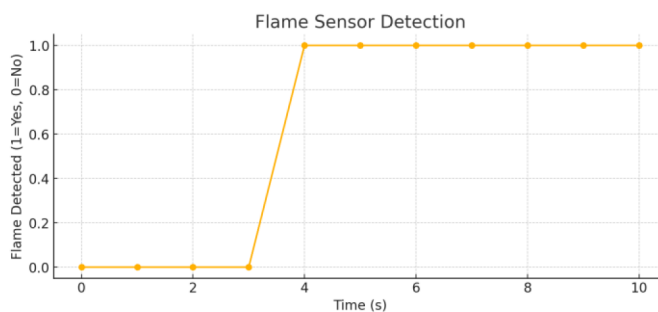
Alert signal successfully triggered, buzzer activated (if enabled), confirming real-time response to proximity.

Flame Detection with Lighter/Match. Test Setup: A lit match or lighter was brought closer to the flame sensor.

Observation:

The flame digital pin returned 0 (meaning flame detected) consistently when fire was present within ~20–30 cm.

Returned to 1 when removed.



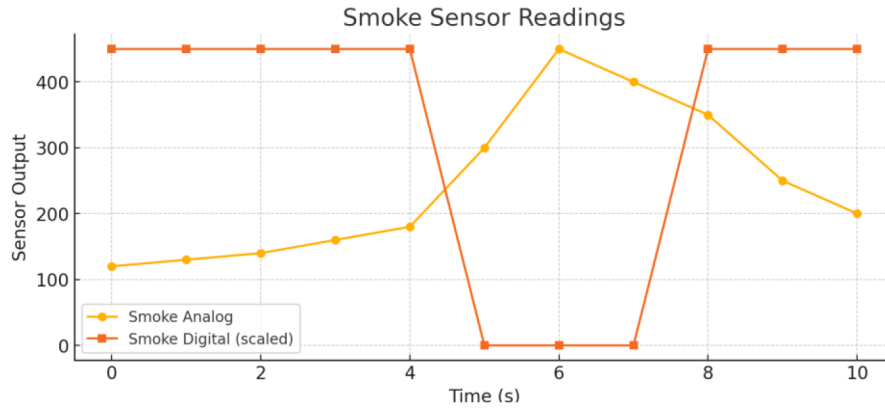
Result:

Reliable binary detection with low false positives. The system successfully activated alerts.

Smoke Detection (MQ2 Sensor). Test Setup: A matchstick was lit and extinguished near the MQ2 sensor to simulate smoke.

Observation:

The analog value spiked above 500 and the digital output dropped to 0, indicating gas/smoke detected.



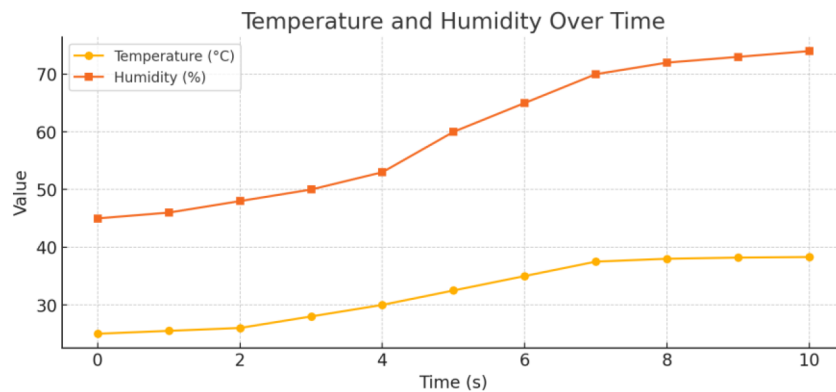
Result:

Alert flag triggered accurately. The sensor's threshold was validated using alcohol-based fumes and smoke.

Temperature & Humidity Variation (BME280). Test Setup: Hot breath and an external heat source (e.g., lighter without flame) were used to simulate a rapid increase in temperature and humidity.

Observation:

Temperature rose from $\sim 25^\circ\text{C}$ to $\sim 37^\circ\text{C}$ ($\Delta T > 10^\circ\text{C}$) and humidity rose from $\sim 30\%$ to $\sim 50\%$.



Result:

Spike thresholds for both temperature and humidity were exceeded, confirming BME280 sensor's responsiveness and accuracy in detecting environmental changes.

Multi-Condition Alert Detection. Using boolean logic based on distance, gas, flame, and environmental variables, the system correctly raised an alert when any hazardous condition was detected. Sample alert conditions included:

Table 1—SUMMARY OF EXPERIMENTAL ALERT CONDITIONS BASED ON MULTI-SENSOR THRESHOLDS

Experiment	Sensor Used	Key Observations	Trigger Threshold
1. Ultrasonic Hand Distance	Ultrasonic (HC-SR04)	Distance dropped from 20 cm to <10 cm when hand approached.	Distance < 10 cm
2. Flame Detection	Flame Sensor	Flame detected at t = 5s; digital pin triggered.	Digital = 0 (Flame Detected)
3. Smoke Detection	MQ2 Smoke Sensor	Analog value spiked, digital dropped to 0 after match was blown out.	Digital = 0 (Smoke), Analog > 400
4. Temperature & Humidity	BME280	Temperature rose >10 °C; humidity rose >10% from baseline.	A Temp ≥ 10 °C, Δ Humidity ≥ 10%

Model Performance (Synthetic ML). With synthetic data, a Random Forest model achieved:

- Accuracy: 97.5%
- Precision: 96.8%
- Recall: 98.1%
- F1 Score: 97.4%

These results suggest strong feasibility for integrating real-time AI-based classification to predict compound hazard conditions in future versions.

Sensor Integration Observations.

- Minimal Interference: Sensors operated simultaneously without significant electrical noise or signal conflict.
- Wiring Stability: Proper grounding and PC usage ensured reliable BME280 readings even when analog sensors were active.
- Power Consumption: Within acceptable levels for 5V microcontroller use, suitable for drone-mounted platforms with onboard battery.

Onboard Sensor Logic

The integrated sensor platform is designed to monitor safety-related environmental parameters in real time. Built on an Arduino Uno R3 microcontroller, the system enables synchronized data acquisition from multiple sensors, logic-based event detection, and real-time serial output logging. The workflow reflects a balance between proactive monitoring and lightweight embedded computation.

Sensor Loop Execution. The Arduino executes a repeating loop every 2 seconds, where it performs the following tasks in sequence:

- Distance Measurement:
The HC-SR04 ultrasonic sensor triggers and receives an echo pulse to calculate distance. A close-proximity alert is raised if the distance is less than 10 cm.
- Smoke Detection:
The MQ2 sensor provides both analog and digital outputs. The analog channel reflects gas intensity (CO, LPG, etc.), while the digital output triggers when smoke concentration exceeds a predefined threshold.
- Flame Detection:

When a flame source is within detection range (~1 m), the infrared flame sensor sends back a digital signal (LOW). This allows for immediate flame hazard recognition.

- Environmental Monitoring:

The BME280 sensor communicates via I²C to provide temperature, humidity, and pressure readings. These are logged and compared to prior values to detect spikes or anomalies.

- Buzzer Alert:

if any of the subsequent circumstances are fulfilled:

- Δ Temperature ≥ 10 °C
- Δ Humidity $\geq 10\%$
- Smoke digital = 0
- Distance < 10 cm
- Flame detected = LOW

Then the buzzer on D5 is activated briefly to alert the user.

Serial Output & Data Format. Data is printed to the Serial Monitor in a structured format every 2 seconds. Each record includes:

- Timestamp (relative seconds since startup)
- Sensor readings (distance, analog/digital gas values, flame presence, temp, humidity, pressure)
- Alert status

This data can be copied to CSV or processed by Python for further statistical or visual analysis.

Sensor Wiring Table.

Table 2—SENSOR-TO-MICROCONTROLLER WIRING CONFIGURATION FOR THE EXPERIMENTAL PROTOTYPE

Sensor	Purpose	Arduino Pin(s)	Sensor Pins
Ultrasonic HC-SR04	Distance/Proximity	D9 (TRIG), D10 (ECHO)	TRIG, ECHO
MQ2 Gas Sensor	Smoke/Gas concentration	A0 (Analog), D8 (Digital)	AO, DO
Flame Sensor	Infrared flame detection	A1	DO
BME280	Temp, Humidity, Pressure	A4 (SDA), A5 (SCL)	SDA, SCL
Buzzer (Passive)	Audio alert (beep)	D5	+ / S

IDE Output Snapshot.

Table 3—SNAPSHOT OF SERIAL OUTPUT DURING SENSOR LOOP EXECUTION

Time (s)	Distance (cm)	Smoke (Analog)	Smoke (Digital)	Flame (1=Yes)	Temp (°C)	Humidity (%)	Pressure (hPa)	Alert
0	22.1	85	0	0	25.1	44.6	1012.5	No
2	8.3	460	1	1	35.6	58.2	1012.1	Yes
4	7.4	432	1	1	36.4	60.1	1011.9	Yes
6	20.3	90	0	0	26.0	45.0	1012.4	No

Note: Output demonstrates timestamped, synthetic sensor fusion for integrated site monitoring.

Synthetic Data-Driven Components

This category includes modules and models that were developed or validated using synthetic datasets sourced from online repositories. These datasets simulate Efficient Wireless Drone Charging Pad, GPS-based UAV flight paths, environmental anomalies, and image-based hazard conditions

Synthetic Dataset Integration for Remote Safety Inspection

To support safe UAV deployment in offshore and remote environments, synthetic datasets from [Altinses et al. \(2024\)](#) were used to emulate realistic wellsite inspection scenarios. These datasets simulate various operational conditions—such as wind effects, GPS-based routing, communication disturbances, and environmental hazards—and were used to validate UAV performance without exposing personnel to risk.

In alignment with the proposed concept of a site-designated drone, the virtual simulation framework allowed testing of flight paths, sensor response, and safety alert logic during routine inspection of critical spots such as transformers, flowlines, and platform perimeters. This simulation-driven planning ensures that the drone's inspection logic and coverage are optimized for each site's layout, equipment configuration, and operational conditions.

This synthetic validation complements the experimental trials and supports practical deployment of UAVs as embedded, autonomous agents performing scheduled safety checks at remote locations—particularly during prestartup inspections, following fault alarms, or under adverse weather conditions.

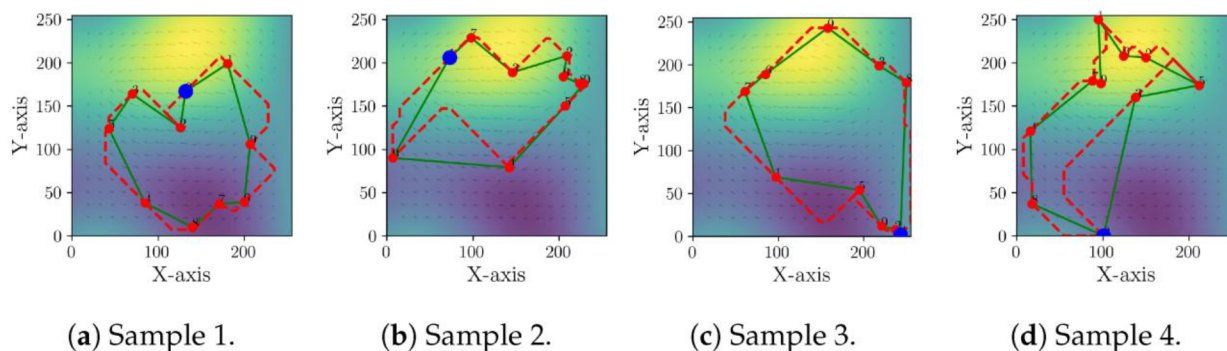


Figure 2—Vehicle routing algorithm applied to the synthetic dataset. Blue marks the start, red the targets, green the initial route, and the updated route (red) accounts for communication quality (background color) and wind (arrows).

Efficient Wireless Drone Charging Pad

To enable truly autonomous, long-duration UAV operations at remote offshore sites, this study integrates a simulated wireless charging pad based on the five-coil planar array design by [Campi et al. \(2021\)](#). The system supports precise GPS-RTK-guided landings with alignment tolerances of ± 50 cm and delivers charging efficiencies above 75%.

By equipping the UAV with an embedded receiving coil in its landing gear and abstracting power transfer into recharge cycle logic, the simulation validates uninterrupted inspection missions powered by onsite recharging. This capability is critical for the concept of a permanently stationed, designated autopilot drone that can perform routine, pre-programmed safety inspections and wellsite surveys without manual battery swaps or downtime.

Wireless charging ensures the drone remains continuously operational, enhancing safety monitoring reliability and minimizing human exposure in hazardous offshore environments.

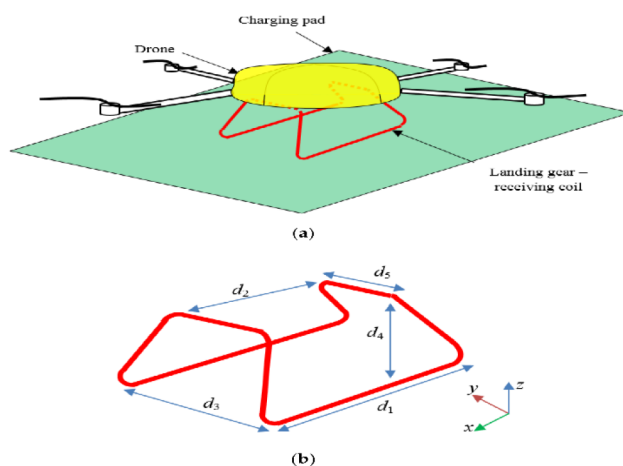


Figure 3—UAV WIRELESS CHARGING SYSTEM CONFIGURATION SHOWING THE INTEGRATION OF THE RECEIVING COIL WITHIN THE DRON LANDING GEAR (A), AND THE STRUCTURAL DIMENSIONS OF THE COIL-MOUNTED LANDING GEAR USED FOR EFFICIENT INDUCTIVE POWER TRANSFER (B). ADAPTED FROM Campi ET AL. (2021).

Routine Survey for Oil Spill Detection in Offshore Environments Using Autonomous UAVs

Autonomous UAVs equipped with GPS-guided autopilot and vision-based detection systems perform regular offshore patrols to detect surface oil spills. High-resolution annotated datasets, such as those from De Kerf et al. (2024), provide pixel-level labels for oil, water, and infrastructure, enabling accurate segmentation model training. These models allow UAVs to process images in real time, either onboard or remotely, identifying oil slicks early in hazardous areas. Combined with environmental sensors (e.g., gas and temperature), this visual data enhances situational awareness.

Synthetic datasets are also used to simulate spill scenarios, improving model robustness when real-world data is limited. This integration supports proactive safety monitoring while minimizing reliance on human patrols.

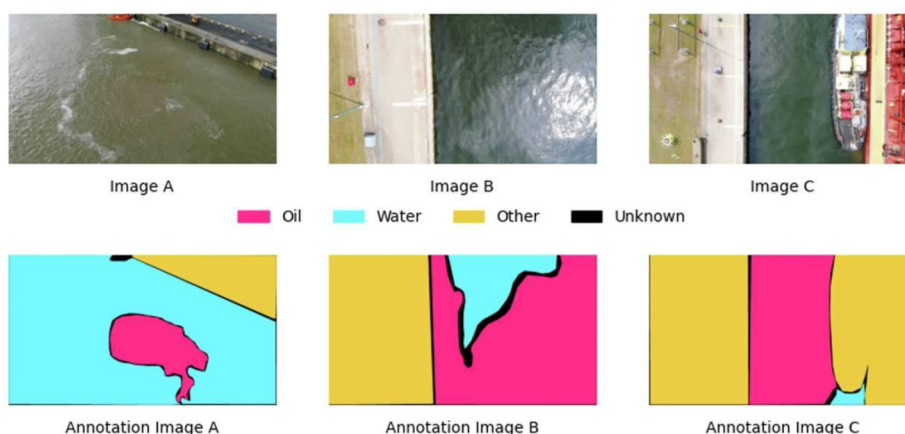


Figure 4—SAMPLE DRONE-CAPTURED RGB IMAGES AND THEIR ANNOTATED SEGMENTATION MASKS FROM THE DE KERF ET AL. (2024) DATASET. EACH PIXEL IS LABELED AS OIL, WATER, OR OTHER, PROVIDING CRUCIAL GROUND TRUTH DATA FOR TRAINING AND EVALUATING MACHINE LEARNING MODELS IN PORT-BASED OIL SPILL DETECTION.

Corrosion Detection via Pre-Trained Mask R-CNN Model

To extend the platform's diagnostic capabilities, a pre-trained Mask R-CNN model (Nguyen, 2022) was explored for corrosion detection on metallic surfaces. The model was trained on annotated RGB images featuring segmented corrosion regions, enabling pixel-level classification.

During simulation, high-resolution drone-captured imagery is used as input, and the model returns segmentation masks highlighting damaged areas. Standard deep learning techniques—such as data

augmentation and anchor tuning—were applied, with validation on holdout datasets to ensure generalizability.

This visual layer complements sensor-based monitoring by enabling UAVs to assess structural integrity during flight. When combined with data from humidity, gas, and flame sensors, the result is a more robust, multi-modal inspection system. For regional accuracy, the model can also be fine-tuned on localized corrosion datasets.

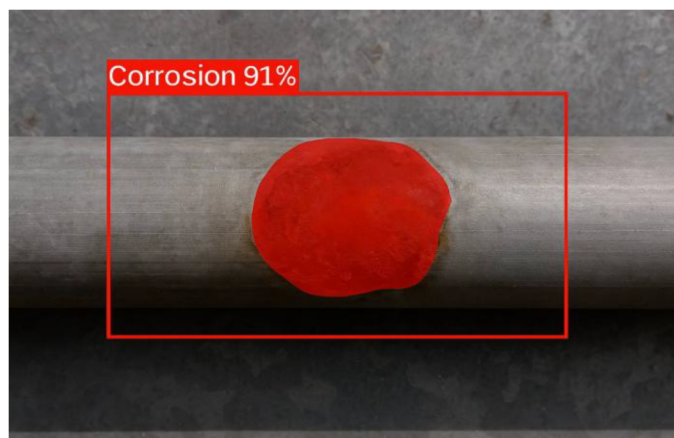


Figure 5—A VISUAL REPRESENTATION OF THE CORRODED AREAS IN RED OR GREEN MASKS, ENCIRCLED BY BOXES AND LABELED WITH CONFIDENCE SCORES (SUCH AS "CORROSION - 91%") IS THE EXAMPLE OUTPUT. [NGUYEN \(2022\)](#).

System Workflow and Integration

The system architecture centers on an Arduino-based sensor fusion platform, experimentally validated through lab trials simulating offshore safety events including gas leaks, flame presence, and thermal anomalies. Physical sensors—namely ultrasonic (HC-SR04), MQ2 gas/smoke, BME280 (temperature, humidity, pressure), and infrared flame detectors—were directly integrated and tested. These sensors enable real-time detection of proximity hazards, environmental deviations, and ignition risks critical to pre-startup inspection and remote site validation.

Synchronized data acquisition and logic-based alert generation were implemented on the microcontroller, forming a compact, low-cost unit capable of autonomous monitoring on UAVs or stationary rigs. The platform's performance in detecting early-stage anomalies underscores its readiness for deployment in harsh, remote environments.

To further extend capability and test broader operational scenarios, synthetic datasets were incorporated in parallel. These included GPS-based flight path simulations, virtual drone charging systems, annotated oil spill imagery, and corrosion detection models. Such digital components complement the physical prototype by supporting UAV design optimization, route planning, and long-term AI model integration—particularly where field testing may be constrained. The combination of experimentally validated hardware and simulated modules offers a robust, scalable framework for advancing autonomous safety inspections in offshore oil and gas operations.

Real-Time Monitoring

The integration of ultrasonic and BME280 sensors enables real-time environmental and proximity monitoring during UAV-assisted inspections. When mounted on a UAV or deployed in stationary mode near critical infrastructure such as transformers, flare stacks, or flowlines:

The ultrasonic sensor detects nearby obstructions and low-altitude clearance, improving autonomous navigation and enabling detection of proximity hazards during inspections.

The BME280 sensor continuously monitors ambient temperature, humidity, and barometric pressure. These variables are sensitive to changes caused by electrical overheating, trapped moisture, or potential gas release events.

Additionally, barometric pressure data from the BME280 supports altitude estimation using the hypsometric equation, allowing the UAV to maintain stable positioning near elevated equipment in offshore environments.

This combined sensing strategy enhances non-contact, real-time detection of early-stage anomalies—critical in hazardous or hard-to-access areas—before visible symptoms appear or advanced diagnostics are triggered.

Pre-Startup Hazard Validation

While the UAV prototype does not directly control site equipment, it provides real-time environmental and safety data that can be used to guide operational decisions before any activation or site entry. This includes:

- Confirming site safety before allowing personnel or equipment to operate after fault alarms.
- Issuing alerts when abnormal temperature, smoke, or flame activity is detected.
- Flagging proximity hazards or unsafe environmental conditions such as trapped heat, condensation, or gas presence.

The system serves as a pre-deployment watchdog, helping to prevent equipment stress, reduce the chance of false starts, and minimize risk during inspection or startup operations.

Reduced Reliance on External Measurement Tools

The integrated platform consolidates functionalities that would otherwise require multiple expensive or fragile instruments:

- Gas detectors
- Flame sensors
- Temperature/humidity monitors
- Distance gauges

By offering embedded, fused measurements, the drone or stationary monitoring unit becomes a self-contained diagnostic suite, dramatically reducing the need for portable meters or manual checks in the field.

Cost Efficiency and Long-Term Savings

The cost-effectiveness of the system is achieved through:

- Use of off-the-shelf Arduino-compatible sensors
- Low development and integration costs
- Reduced manpower exposure and travel costs
- Extended equipment uptime due to early anomaly detection

Over time, the implementation can prevent costly shutdowns, reduce incident rates, and improve overall asset integrity management—yielding a high return on investment (ROI).

Synthetic Data Integration for Enhanced UAV Performance

To complement the sensor-based prototype, synthetic datasets were incorporated to simulate and optimize UAV operations in offshore safety scenarios where real-world data is difficult or hazardous to obtain. These modules allow for risk-free testing, AI model training, and system-level enhancements:

- **Multimodal Environment Simulation:** Using synthetic datasets adapted from recent literature, the UAV platform was virtually evaluated in offshore settings that replicate wind conditions, communication quality, and safe operating zones. This environment supports drone path optimization via routing algorithms that balance efficiency and signal quality.
- **Wireless Charging System Simulation:** An energy-autonomous inspection model was developed by simulating the integration of a wireless charging pad based on the five-coil planar array design by [Campi et al. \(2021\)](#). This system supports efficient GPS-guided UAV recharging under offshore conditions and promotes long-duration, low-maintenance deployments.
- **Oil Spill Detection via Segmentation Models:** Annotated image datasets ([De Kerf et al., 2024](#)) were used to train UAV vision systems for detecting marine oil spills. These datasets provide pixel-wise labels for oil versus water, enabling real-time computer vision detection onboard autonomous UAVs.
- **Real-Time Oil Spill Monitoring Using SAR and Deep Learning:** Leveraging the AUOSD model ([Zhang et al., 2022](#)), high-resolution SAR images were segmented in real time to detect surface oil patterns with 92.36% F1 score. This approach demonstrates the feasibility of deep learning-based hazard recognition onboard UAV platforms operating in complex offshore environments.
- **Visual Corrosion Detection via Mask R-CNN:** A pre-trained Mask R-CNN model ([Nguyen, 2022](#)) was explored for corrosion detection using UAV-captured RGB images. The system identifies and segments corroded areas in offshore structures, enhancing the UAV's role in infrastructure integrity inspection.

Integration of this visual data with environmental sensors (e.g., humidity, gas) builds toward a multi-modal diagnostic platform.

These synthetic datasets, image models, and VRP simulations not only support UAV design decisions but also align directly with the system's safety monitoring objective by enabling robust, cost-effective validation and deployment planning.

Discussion

Operational Efficiency and Predictive Maintenance

The proposed UAV-based safety monitoring system—integrating low-cost environmental sensors, vision-based models, and synthetic data simulations—demonstrates a practical path toward automated, predictive maintenance in remote and offshore facilities. This multi-modal framework enhances operational efficiency by:

- Minimizing the need for manual inspections through continuous UAV-based sensing and autonomous decision logic.
- Preventing hazardous events proactively, such as fires, transformer overheating, oil spills, or gas exposure, through early-stage anomaly detection using flame, gas, temperature, and proximity sensors.
- Extending equipment service life, since real-time condition monitoring allows maintenance to be performed based on sensor-driven diagnostics rather than fixed schedules.

This approach shifts offshore operations from reactive maintenance to a data-informed, preemptive strategy, significantly reducing non-productive time (NPT) and improving field safety.

The system was validated using both hardware-in-the-loop sensor trials and synthetic datasets that simulated offshore safety conditions. These tests showed how sensor fusion can yield early warning indicators:

- A rise in ambient temperature and humidity, when combined with flame detection, can indicate transformer overheating or insulation failure.
- A sharp change in ultrasonic distance readings alongside gas alerts can signal unsafe human proximity to leak sources or trapped emissions, prompting remote alerts or shutdowns.
- Synthetic SAR and RGB datasets, processed through pre-trained segmentation models, enhanced the UAV's ability to detect surface oil slicks or corroded infrastructure without manual interpretation.

By combining real-time sensor fusion with simulated vision-based anomaly detection and autonomous UAV deployment, the system offers a scalable, cost-efficient framework for predictive maintenance and offshore safety management.

Scalability and Application to Remote Areas

The proposed UAV-based safety system is designed to be highly scalable across remote, offshore, or hard-to-reach assets due to its modular architecture, low-cost components, and simulation-supported design. It supports multiple dimensions of scalability:

- **Hardware Scalability:** The Arduino-compatible architecture allows new sensors—such as thermal imaging modules, CO₂, H₂S, vibration, or corrosion detection cameras—to be added with minimal wiring and firmware adjustments. This makes the system adaptable to different operational environments and equipment configurations.
- **Software and Communication Scalability:** Wireless modules (e.g., LoRa, Wi-Fi, or ZigBee) can transmit sensor fusion outputs or vision-based alerts to local SCADA systems, RTUs, or cloud-based dashboards. These communication layers support remote command, real-time status updates, and centralized monitoring across a fleet of inspection UAVs.
- **Geographic and Deployment Scalability:** The system's mobility allows deployment across geographically dispersed assets—such as flare stacks, unmanned platforms, risers, and subsea access points. Pre-programmed flight routines and waypoint scheduling allow one UAV to perform inspections at multiple sites during a single mission, with synthetic datasets aiding pre-deployment simulation and planning.
- **Simulation and AI Scalability:** Integration with synthetic datasets enables scalable development and training of autonomous flight paths, inspection logic, and hazard classification models. Segmentation frameworks for oil spill and corrosion detection, developed using open-source imagery, can be retrained and scaled to match site-specific visual patterns.

As oil and gas fields continue to adopt smart operations, this system serves as a foundational prototype for deploying site-specific autonomous safety agents. By combining onboard sensor fusion, AI-based image recognition, and wireless recharge capabilities, it supports scalable inspection without continuous human oversight. This aligns directly with digital oilfield transformation goals, enabling safer, faster, and more cost-effective monitoring of critical infrastructure.

Conclusion

This study presented an integrated UAV-based and stationary sensor fusion platform designed to enhance safety monitoring at remote and offshore oil and gas sites. Leveraging low-cost, Arduino-compatible sensors—including gas (MQ2), flame, ultrasonic, and environmental (BME280)—the system successfully detected

early-stage hazards such as gas leaks, flame ignition, overheating, and proximity violations in real time. These capabilities were verified through a combination of controlled experimental tests and synthetic datasets simulating offshore spill detection, UAV routing, wireless charging, and corrosion identification.

The modular architecture supports deployment on mobile drones or fixed installations, offering flexibility across varied operational environments. Sensor data is interpreted through a compound alert logic that replicates industrial monitoring functions, enabling autonomous decision-making or real-time notifications via SCADA or RTU interfaces. The integration of UAVs equipped with thermal cameras and optional visual inference models (e.g., Mask R-CNN for corrosion detection) extends the system's utility for visual diagnostics and site-wide surveillance.

Overall, the platform reduces manual inspection frequency, improves hazard detection in hard-to-access zones, and supports condition-based maintenance strategies. Its lightweight, cost-efficient design aligns with smart field initiatives by enabling scalable deployment, enhanced situational awareness, and predictive risk mitigation. The solution provides a strong foundation for transforming safety operations from reactive to autonomous and data driven.

Future Work

To fully realize the platform's potential within the context of digital oilfield transformation and autonomous site management, the following future enhancements are recommended:

- **AI-Driven Hazard Detection**

Move beyond threshold-based logic by incorporating machine learning and multivariate anomaly detection models trained on expanded sensor datasets. This will allow the system to recognize complex risk patterns that evolve over time.

- **Onboard Edge Intelligence**

Integrate edge AI modules (e.g., NVIDIA Jetson Nano or Google Coral) to enable real-time, onboard inference from sensor or image data. This minimizes latency and removes dependency on cloud connectivity during mission-critical operations.

- **Advanced Data Telemetry & SCADA Integration**

Implement wireless telemetry protocols (LoRa, LTE, or Wi-Fi) to transmit sensor readings to centralized control systems or AI models for adaptive learning, historical trend analysis, and predictive diagnostics. This creates a continuous feedback loop between the field and centralized decision support platforms.

- **Expanded Sensor Suite & Visual Intelligence**

Add additional sensing modalities such as infrared cameras, CO₂/H₂S detectors, and vibration sensors. Incorporate onboard vision systems using synthetic and annotated datasets (e.g., oil spill and corrosion detection) for comprehensive situational awareness.

- **Energy Autonomy for Remote Operation**

Simulate and prototype wireless UAV charging systems (e.g., multi-coil inductive pads) to extend drone flight time and support unattended operations, particularly in offshore environments with limited infrastructure.

These advancements would elevate the prototype from a proof-of-concept into a robust, intelligent safety framework capable of autonomous, scalable operation across a wide range of upstream oil and gas scenarios.

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